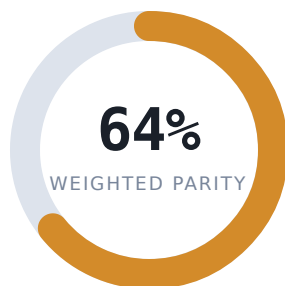


Mirza Bot — PHP → Python Parity

Static-analysis audit of the rewrite/python branch against main (production PHP). Baseline = source of truth.

```
RE-AUDIT post tranche-1 · static
rewrite head c7e0fee
parity 55% → 64%
features scored 96
2026-08-25
```



After tranche-1, the money path works: a customer can pay via a gateway and receive their config, and wallet top-ups credit instead of crashing.

Parity moved **55% → 64%**; the critical block went from **0/4** to **≈3.5/4**. Verified by re-reading every fix in code. Two residuals remain and the revenue-retention features (customer reminders, discounts, referrals) are still tranche-2.

✓ Tranche-1 accepted · not yet production

37

✓ Done / improved

42

⚠ Partial

5

🐛 Wrong / buggy

3

🔒 Risky

9

✗ Missing

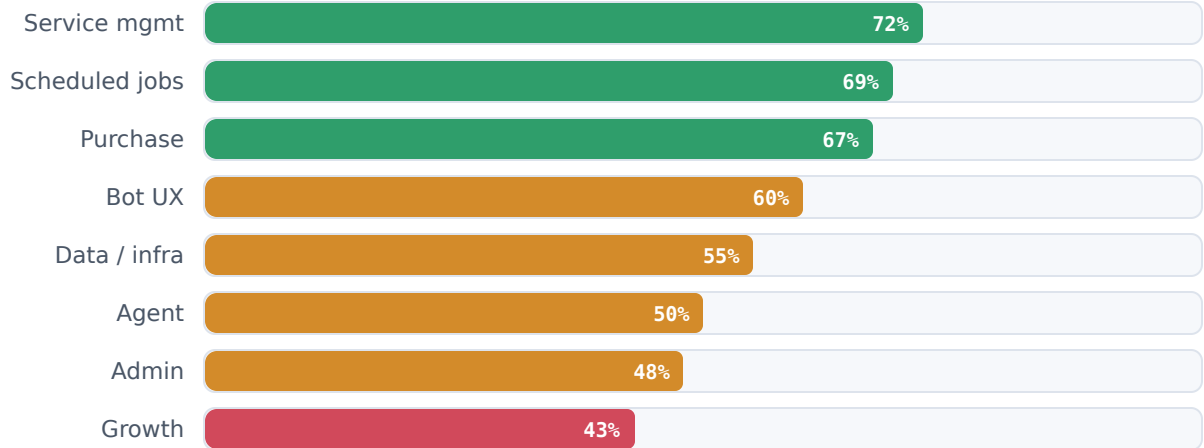
STATUS BREAKDOWN



✓ Done / improved	37	39%
⚠ Partially done	42	44%
🐛 Done but buggy	5	5%
🔒 Done but risky	3	3%
✗ Missing	9	9%

PARITY BY FUNCTIONAL AREA





TOP LAUNCH-BLOCKING RISKS

HIGH

Wallet debit is not race-safe

Charge-before-provision is fixed, but WalletService.change is read-modify-write (no row lock / atomic UPDATE) — two concurrent wallet buys can still double-spend on PostgreSQL.

C4 · F-005

HIGH

Expiry reminders miss the customer

Renewal/volume warnings still post to an admin channel, not the buyer — the renewal engine never reaches paying users (tranche-2).

F-076 · F-077

HIGH

Migrations still a create_all shim

Initial migration loops model metadata instead of real DDL; no compatibility view for the redesigned settings table (tranche-2).

F-094

HIGH

Admin surface still a fraction

~25 commands vs ~177; in-bot text customization and per-panel/product editing still absent — the largest remaining block.

M9

RESOLVED

Gateway settlement (was critical)

RESOLVED in tranche-1: settle_wallet_topup restored, gateway purchases set invoice_id, buyer/admin notified for 5 gateways (cubepay notify still open).

C1 · C2 · C3

AREA DETAIL — STATUS MIX

Purchase & provisioning

67%

Service management

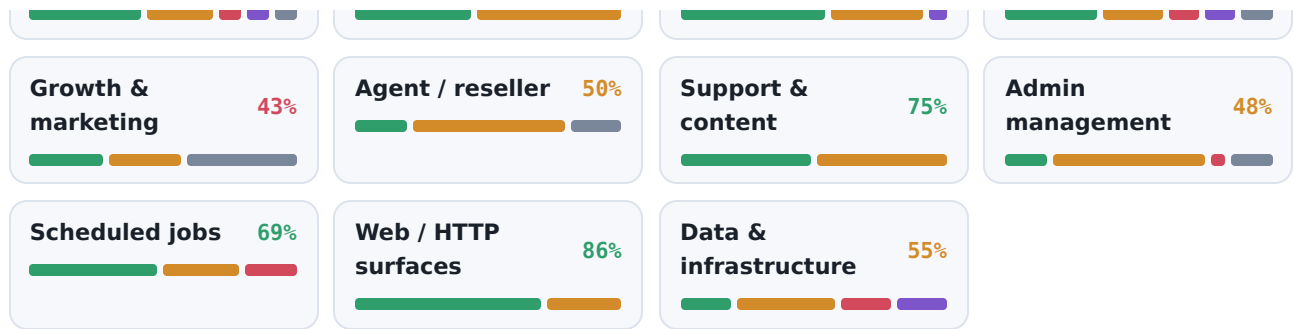
72%

Payments & wallet

82%

Bot UX & onboarding

60%



How completion is calculated. Each of 96 parity features is scored by fidelity to baseline — Done/Improved = 1.0, Risky = 0.7, Partial = 0.5, Buggy = 0.2, Missing = 0.0 — and averaged. Additions in the rewrite (ledger, idempotency, webhook-secret, encryption, plugin registry) are excluded from the denominator. The score measures how much surface area is present, **not** operational readiness of the revenue path.

Re-audit (post tranche-1): the three critical payment defects are fixed and now covered by genuine money-path tests — verified by re-reading every change in `rewrite/python@c7e0fee`. Two residuals remain: the wallet debit is not race-safe (`change()` is read-modify-write, not an atomic conditional decrement), and cubepay's `settle_tuple` path still doesn't notify the buyer or report. · Scope: static pass only — no code executed. Executing the 45-test suite, panel-adaptor round-trips and end-to-end gateway flows remain for the dynamic pass.